

Mathematics for Social Scientists

Q&A and Exam Preparation

Carlos Gueiros

University of Mannheim

Fall 2025

Outline

Exam Instructions

Extra: Random Variables

Exercise: Probability

Probability

Recap: Eigenvectors

Exam Instructions

Logistics:

- **Date & Time:** Wednesday, December 17th, 10:00 a.m. – 12:00 p.m.
- **Location:** A5, B143

What to Bring:

- Pen, non-programmable calculator, and student ID
- One-sided A4 *hand-written* cheat sheet (one page only)

Exam Structure:

- True/False, Multiple Choice, Analysis & Calculus, Linear Algebra, Intro to Probability
- Format similar to the mock exam

Study Tips:

- Content covers: slides, problem sets, mock exam, etc.
- Be strategic – answer what you know first!
- *matlib* R package - Matrix Functions for Teaching and Learning Linear Algebra and Multivariate Statistics

Outline

Exam Instructions

Extra: Random Variables

Exercise: Probability

Probability

Recap: Eigenvectors

Random Variable I

Definition 1 (Random Variable).

Let Ω be the sample space for an experiment. A real-valued function that is defined on Ω is called a **random variable**. The set of values the variable might take is the **distribution** of the random variable.

From Random Variables to Distributions

Probabilities and events

- Probability P assigns numbers to events $A \subseteq \Omega$.
- Once we define a random variable $X : \Omega \rightarrow \mathbb{R}$, we can form new events based on the values of X :

$$\{X = 0\}, \{X = 1\}, \{X \leq 1\}, \{1 < X \leq 2\}, \dots$$

- Each of these is still just a subset of Ω , so it has a probability via P .

Example: two coin tosses

$$\Omega = \{HH, HT, TH, TT\}, \quad X = \text{"number of heads"}.$$

x	event	$P(X = x)$
0	$\{TT\}$	1/4
1	$\{HT, TH\}$	2/4
2	$\{HH\}$	1/4

Distribution of a random variable

- For each possible value (or range) of X we can form an event and compute its probability, e.g.

$$P(X = 0), P(X = 1), P(X \leq 1), P(1 < X \leq 2).$$

- The **distribution of X** is the function that collects all these probabilities for the numerical values of X .
- For a discrete X this is a *probability mass function* (pmf) $p(x) = P(X = x)$.
- For a continuous X it is a *probability density function* (pdf) $f(x)$, where probabilities of intervals come from areas under the curve.

A probability distribution provides a complete summary of the probabilities of all possible values that the random variable can take.

Discrete Random Variables

Discrete random variable

- A random variable X is **discrete** if it can take only a finite or countably infinite number of distinct values. (Examples: number of wars per year, result of a die roll.)

Probability mass function (pmf)

- The distribution of a discrete r.v. is a **probability mass function** (pmf).
- For a discrete X , the pmf is

$$p(x) = P(X = x),$$

assigning probabilities to each possible value x such that

$$0 \leq p(x) \leq 1, \quad \sum_x p(x) = 1.$$

Example (fair die)

$$p(y) = P(X = y) = \begin{cases} 1/6, & y = 1, \dots, 6, \\ 0, & \text{otherwise.} \end{cases}$$

Cumulative distribution function (cdf)

- For a discrete r.v. X , the cdf is

$$F(x) = P(X \leq x),$$

the probability that X is less than or equal to x .

- Equivalently,

$$F(x) = \sum_{i \leq x} p(i).$$

Properties of a cdf

- $F(x)$ is non-decreasing in x .
- $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.
- $F(x)$ is right-continuous.

Note: $P(X > x) = 1 - P(X \leq x) = 1 - F(x)$.

Continuous Random Variables

Continuous random variable

- A random variable X is **continuous** if it can take values on an interval of the real line (uncountably many possible values). (Examples: age, income, temperature.)

Probability density function (pdf)

- The distribution of a continuous r.v. is described by a **probability density function** (pdf) $f(x)$.
- $f(x)$ satisfies

$$f(x) \geq 0 \quad \text{for all } x, \quad \int_{-\infty}^{\infty} f(x) dx = 1.$$

- Probabilities of intervals are areas under the curve:

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

Cumulative distribution function (cdf)

- For a continuous r.v. X with pdf f , the cdf is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt.$$

- As before, $F(x)$ is non-decreasing, $\lim_{x \rightarrow -\infty} F(x) = 0$, $\lim_{x \rightarrow \infty} F(x) = 1$.

Link between f and F

- When F is differentiable,

$$f(x) = F'(x).$$

- For a continuous r.v. X , $P(X = x) = 0$ for any single point x .

Outline

Exam Instructions

Extra: Random Variables

Exercise: Probability

Probability

Recap: Eigenvectors

Probability: Meteor Test

Problem.

Only 1 in 100 meteors that come close to Earth actually hits the planet.

A test correctly classifies:

- 80% of meteors that will hit Earth (sensitivity),
- 90% of meteors that will not hit (specificity).

For a new meteor, the test returns a positive result (“will hit”). A friend claims this means the meteor will hit with probability $> 50\%$.

Do you agree? Justify your answer (show your work).

Hint:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c).$$

Probability: Meteor Test – Solution

Goal

We want

$$P(\text{hit} \mid \text{test says "will hit"}).$$

Step 1: Define events

$$A = \{\text{meteor hits Earth}\}, \quad A^c = \{\text{meteor does not hit}\},$$

$$B = \{\text{test says "will hit"}\}.$$

Step 2: From story to probabilities

- Only 1 in 100 meteors hits:

$$P(A) = 0.01, \quad P(A^c) = 0.99.$$

- If the meteor will hit, the test says “will hit” 80% of the time:

$$P(B \mid A) = 0.8.$$

- If the meteor will not hit, the test still says “will hit” 10% of the time:

Step 3: Formula we will use

We use Bayes' rule:

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B)}.$$

To apply it, we first need $P(B)$, the probability that the test says “will hit”:

$$P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c).$$

On the next slide we plug in the numbers and simplify.

Probability: Meteor Test – Solution

Step 4: Compute $P(B)$

$$\begin{aligned}P(B) &= P(B \mid A)P(A) + P(B \mid A^c)P(A^c) \\&= 0.8 \cdot 0.01 + 0.1 \cdot 0.99 \\&= 0.008 + 0.099 \\&= 0.107.\end{aligned}$$

Step 5: Compute $P(A \mid B)$

$$\begin{aligned}P(A \mid B) &= \frac{P(B \mid A)P(A)}{P(B)} \\&= \frac{0.8 \cdot 0.01}{0.107} \\&= \frac{0.008}{0.107} \\&\approx 0.0748 \approx 7.5\%.\end{aligned}$$

Step 6: Discussion

Even after we know the test said “will hit”, the chance that the meteor actually hits is only

$$\boxed{\approx 7.5\%},$$

which is far below 50%.

The key reason is that hits are very rare (only 1%). There are many more meteors that *do not* hit, and a small fraction of these still make the test say “will hit”. These false alarms dominate the positive results.

We do *not* agree with the friend's claim that the probability is greater than 50%.

Outline

Exam Instructions

Extra: Random Variables

Exercise: Probability

Probability

Recap: Eigenvectors

Example: eigenvalues and eigenvectors

Matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Step 1: eigenvalues

We compute the characteristic polynomial

$$P_A(\lambda) = \det(A - \lambda I_2) = \det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix}.$$

$$P_A(\lambda) = (2 - \lambda)(2 - \lambda) - 1 = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3.$$

Factor:

$$P_A(\lambda) = (\lambda - 1)(\lambda - 3),$$

so the eigenvalues are

$$\lambda_1 = 1, \quad \lambda_2 = 3.$$

Claim: example eigenvectors are

$$v^{(1)} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad \text{for } \lambda_1 = 1, \quad v^{(2)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{for } \lambda_2 = 3.$$

Check

$$Av^{(1)} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} = 1 \cdot v^{(1)}.$$

$$Av^{(2)} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \cdot v^{(2)}.$$

So the directions $(-1, 1)$ and $(1, 1)$ really are eigenvector directions.

Getting the eigenvectors for $\lambda_1 = 1$

For

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

we find the eigenvectors by solving $(A - \lambda I_2)v = 0$.

Step 1: plug in $\lambda_1 = 1$

$$A - I_2 = \begin{pmatrix} 2-1 & 1 \\ 1 & 2-1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Step 2: solve the homogeneous system

$$(A - I_2)v = 0 \quad \Rightarrow \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This gives one equation:

$$v_1 + v_2 = 0 \quad \Rightarrow \quad v_1 = -v_2.$$

Let $v_2 = a$ (any real number). Then

$$v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} -a \\ a \end{pmatrix} = a \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Getting the eigenvectors for $\lambda_2 = 3$

Again for

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

we solve $(A - \lambda I_2)v = 0$, now for $\lambda_2 = 3$.

Step 1: plug in $\lambda_2 = 3$

$$A - 3I_2 = \begin{pmatrix} 2-3 & 1 \\ 1 & 2-3 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Step 2: solve the homogeneous system

$$(A - 3I_2)v = 0 \quad \Rightarrow \quad \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This gives

$$-v_1 + v_2 = 0 \Rightarrow v_1 = v_2.$$

Let $v_1 = b$. Then

$$v = \begin{pmatrix} b \\ b \end{pmatrix} = b \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$